Photon emission characteristics of avalanche photodiodes

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Abstract. Temporal decay of photon emission from avalanche photodiodes (APDs) is demonstrated. The steep rise of the reverse voltage can cause a higher probability of photon emission. For a silicon avalanche photodiode, the photon emission has a broad spectral distribution from 500 to 1100 nm with two peaks at 750 and 994 nm, respectively. The number of emitted photons increases by 44 Mcount/(s mA sr) as the APD's reverse current increases. This suggests that the reverse current, instead of the reverse voltage, applied to the APD is the main determinant of the photon emission. It is furthermore shown that the distribution of the photon emission is a super-Poisson distribution. [DOI: 10.1117/1.1950087]

Subject terms: avalanche photodiode; temporal behavior; spectral distribution; photon number distribution.

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1 Introduction

Avalanche photodiodes are high-speed, high-sensitivity photodiodes in which internal amplification is achieved by applying a reverse voltage. Compared to standard photomultipliers, they have higher quantum efficiency and are more stable and more robust to external conditions. These advantages have led APDs to a wide variety of applications.[1–7] However, when APDs work in avalanche mode, which triggers an avalanche of hot carriers on absorption of a photon, photons have been found to be emitted from the avalanche region.[8–13] This photon emission may cause some background noise when another detector absorbs the photons, or when the optical system is set up so that photons emitted from a detector can be reflected back on it. Such photon emission has also been investigated in APDs working in Geiger mode for single-photon detection.[8,11–13] All of these works, though, have been focused on photon emission induced by the incident light. Photons emission may also be induced by applying a reverse voltage above the breakdown voltage, as shown in the present work.

In this paper, we investigate the temporal behavior of the photon emission of APDs induced by electroluminescence. The spectral distribution and photon number distribution of this emission are also studied in detail. In the end, some advice is give out for applications of APDs utilizing these particular characteristics.

2 Experimental Setup

Figure 1 shows a diagram of our experimental setup. A single-photon counting module (SPCM, PerkinElmer SPCM-AQR-15) is used to record the emitted photons of an APD (S3884, Hamamatsu). The reverse voltage for the APD is provided by a high-voltage generator. The output of this generator is controlled by a voltage trigger source. When a square wave is applied to the trigger source, a high-voltage square wave is generated to provide the periodic reverse voltage. A digital oscilloscope (Tektronix TDS 1002) is used to monitor the electrical current. Lenses L1 ($f_1 = 100$ mm) and L2 ($f_2 = 30$ mm) are used to focus and collimate photons radiated from the APD in order to improve the detection efficiency. The focused beam is diffracted by a monochromator with a 1200-groove/mm grating blazed at 500 nm and then focused by a lens L3 ($f_3 = 100$ mm). The SPCM is placed at the focal point of L3. The output electrical pulse from the SPCM is recorded by a digital counter (SRS 620, Stanford Research Systems). In order to avoid optical damage and the effects of ambient photons, the APD and the SPCM are placed in an optical shielding box. A computer is used to control the wavelength selection of monochromator and for communication with the SR620 counter. The APD has a breakdown voltage of $V_B = 172$ V (at room temperature) and a sensitive diameter of 1.5 mm. Its spectral response range is 400 to 1000 nm, and its quantum efficiency is about 75% at its central response wavelength $\lambda = 852$ nm. A cooling fin is mounted under the APD to avoid thermal damage. The SPCM has a response range of 400 to 1060 nm and an active diameter

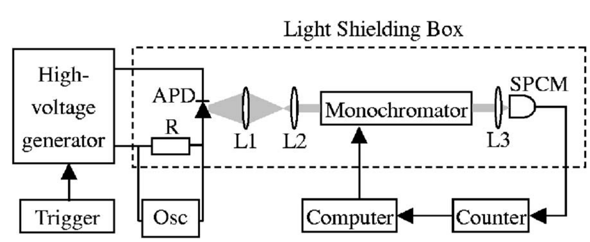

Fig. 1 Simplified diagram of the experimental setup. The bias resistance R is 500 Ω. L1, L2, and L3 are lenses. Osc is a digital oscilloscope. SPCM is a single-photon counting module. The APD and the SPCM are put in an optical shielding box.

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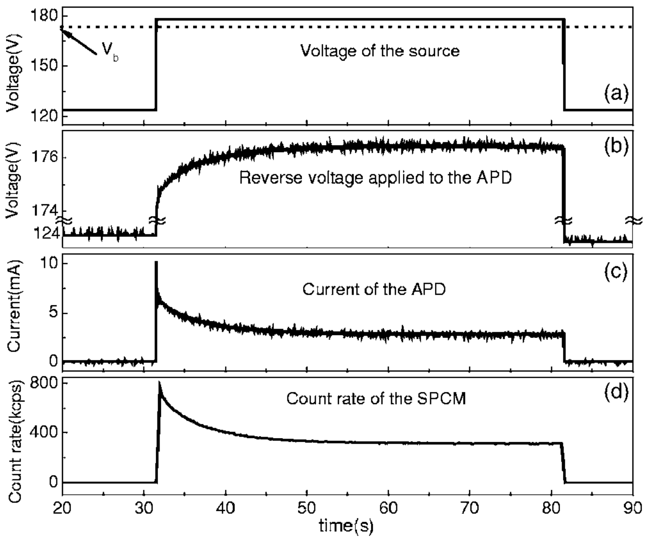

Fig. 2 Temporal traces of the photons emission. (a) The solid curve gives the voltage of the source, and the dotted curve gives the typical breakdown voltage of the APD at room temperature. (b) Reverse voltage applied to the APD. (c) Temporal trace of current of the APD. (d) Temporal trace of count rate of the SPCM with the sample interval T=0.216 s and the bin time τ=0.1 s.

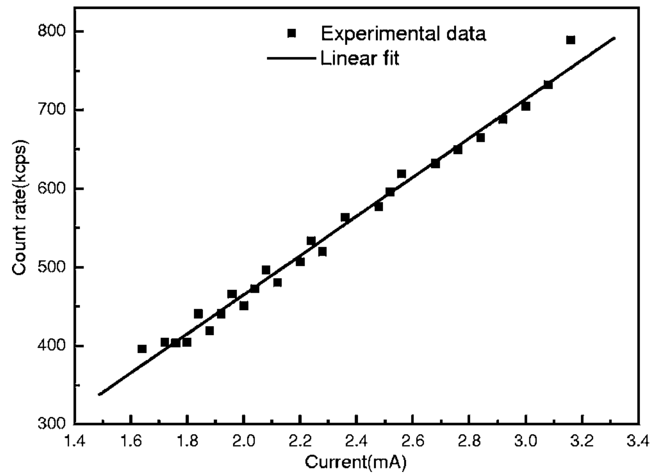

Fig. 3 The photon count rate versus the reverse current of the APD.

of 0.17 mm. The dead time of the SPCM is about 50 ns, and its mean dark count rate is lower than 50 counts/s. The SPCM is both thermoelectrically cooled and temperature-controlled, ensuring stable performance regardless of changes in the ambient temperature.

3 Results and Discussion

3.1 *Temporal Behavior*

A periodic reverse voltage is applied to the APD, with a bias resistance of 500 Ω, to investigate the temporal behavior of the photon emission. Here, the experimental setup is modified slightly. The active areas of APD and SPCM are placed face to face. The current of the APD and the count rate of the SPCM are recorded simultaneously as shown in Fig. 2.

The period of the reverse voltage is 100 s. The low level, about 124 V as shown in Fig. 2(a), is below the breakdown voltage of the APD. The APD does not work at this voltage; the reverse current of the APD is almost zero, and the count rate of the SPCM is due to just the dark counts, shown as the parts before the rising edge and after the falling edge of the source's voltage in Fig. 2(c) and 2(d). The high voltage level of the power source, about 176 V, is 4 V higher than the breakdown voltage of the APD. This voltage triggers the avalanche of the APD, and photon emission occurs. The count rate of the SPCM, as shown in Fig. 2(d), increases and reaches a maximum value simultaneously with the rising edge of the source voltage. Then it gradually decreases to a plateau that lasts till the falling edge of the source voltage. This temporal behavior of photon emission can be fitted with a first-order exponential decay with time constant t=5.15 s. The sample interval time T of the curve in Fig. 2(d) is 0.216 s, and the bin time τ is 0.1 s. The curve of the APD's reverse current, as shown in Fig. 2(c), has almost the same trend as the SPCM's count rate. The maximum count rate in the decay process is 759,410 counts/s, which is 2.4 times of the minimum count rate (314 kcount/s). Furthermore, the maximum reverse current of the APD is 10.24 mA, 3.9 times of the minimum value (2.64 mA). APDs have a higher probability of emitting photons after the steep rise of their reverse voltage.

The relationship between the current and the count rate is shown in Fig. 3. It demonstrates a linear dependence of photon emission on the reverse current. The number of emitted photons, calculated as a function of the spatial angle, is linearly correlated with the APD's current, with a slope of about 44 Mcount/(s mA sr). The current of the APD is the main determinant of the photon emission.

As mentioned, it is unsuitable to record the photoelectric signal from APDs immediately after the steep rise of its reverse voltage. Therefore, the decay certainly will interfere the high-speed detection of APDs operated in the gated mode that is widely used in single-photon detection. To solve this problem, one can decrease the gate frequency to a small value to avoid the long decay process, or keep the APD working in Geiger mode all the time and use external means to perform data selection.[14]

3.2 *Spectral Distribution*

By scanning the wavelength of the monochromator stepwise, the spectral distribution of the photon emission was measured at different reverse currents, and the result is shown in Fig. 4. It is a wide distribution, ranging from 500 to 1000 nm, with a maximum at 750 nm and a weaker peak at 944 nm. The response range of the SPCM constrains the wavelength range. It is shown that the line shapes of the spectral distribution do not change much at different reverse currents, because of the linear relationship between the reverse current and the count rate. When the reverse voltage applied to the APD is below the breakdown voltage, the avalanche process stops, and curve D in Fig. 4(a) shows the corresponding dark counts, which are treated as the background of the spectral distribution measurements.

Considering the quantum efficiency of the APD shown in Fig. 4(b), one can identify a wavelength range

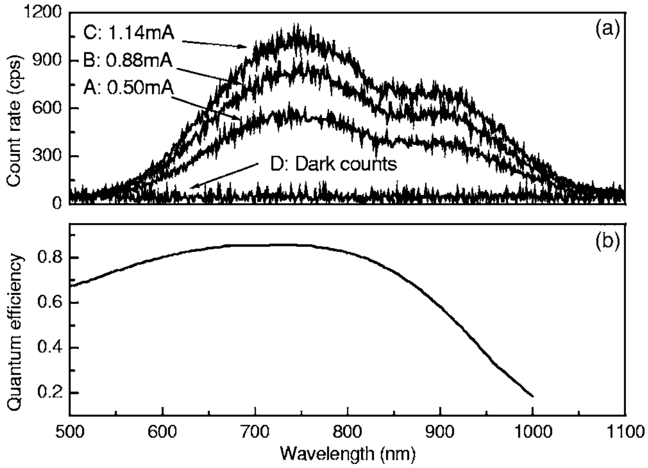

Fig. 4 (a) Spectral distributions of photons emission with different currents of the APD. Curves A, B, C show the spectral distribution at 0.5, 0.88, 1.25 mA, respectively. Curve D is the distribution of dark counts. (b) Quantum efficiency of the SPCM.

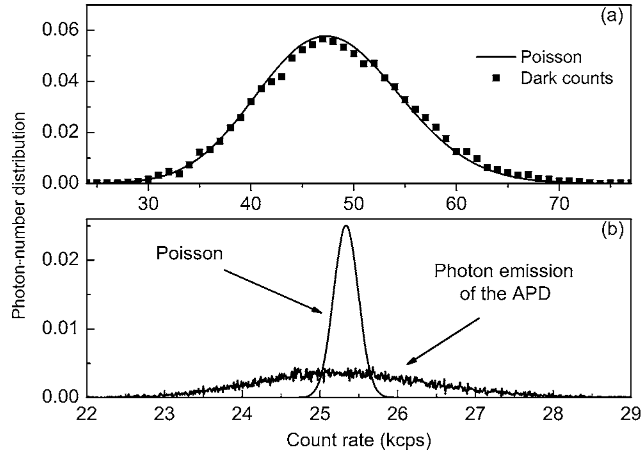

Fig. 5 Photon-number distributions of dark counts and photon emission. (a) Photon-number distribution of dark counts. The distribution of dark counts (dots) agrees with the Poisson distribution (solid curve, with the same mean number as the experimental data). (b) Photon-number distribution of photon emission with the sample interval $T=0.216$ s and the bin time $\tau=0.1$ s. The Poisson distribution curve has the same mean number as the experimental count data of the APD.

(<550 nm) where the APD has sufficient quantum efficiency while no photons are emitted. Therefore, there are two ways to decrease the effects of APD photon emission: one is using a special optical source with central wavelength in this range; the other is adding narrowband interference filters in front of APDs with their central wavelength around the wavelength of the incident light.

3.3 *Photon-Number Statistics*

In order to quantify the fluctuations of photon emission from APD, it is important to compare it with the Poisson distribution

$$P(n)=\frac{\alpha^n e^{-\alpha}}{n!},$$

with α as the mean count rate. Previous work has already shown that a coherent source follows a Poisson distribution, whereas a chaotic source follows a super-Poisson distribution.[14–16]

The dots in Fig. 5(a) show the experimental dark count distribution of the SPCM at room temperature with a sample interval T of 1.116 s, a bin time τ of 1 s, and 15,590 sample points. The photon-number distribution of dark counts can be well fitted with a Poisson distribution with a mean of $\alpha=47$ [solid line in Fig. 5(a)]. This is a similar conclusion to that in Ref. 15.

On the contrary, the photon-number distribution of the APD's photon emission shows a much wider range than the Poisson distribution with a mean of $\alpha=26{,}224$, which is shown in Fig. 5(b). The reverse current of the APD is 1.55 mA. The sample interval T is 0.216 s, and the bin time τ is 0.1 s. The points that have been sampled are as many as 12,608. The result shows that the APD's photon emission has a super-Poisson distribution.

4 Conclusions

In the present work, the photon emission behavior of APDs is investigated systematically. A temporal decay process is observed, which shows that APDs working in gated mode are not suitable to perform high-speed photon detection, as it takes a long time (about 10 s in this experiment) for the APDs' photons emission rate to reach their lowest stable value. Data sample analyses[14] should be performed externally instead of using gated APDs (or SPCMs). The photon emission of silicon APDs has a broad spectral distribution from 500 to 1100 nm; the higher the APD current is, the more photons will be emitted. The photon-number distribution of APDs' photon emission is a super-Poisson distribution. These properties of APDs will be attractive for applications.

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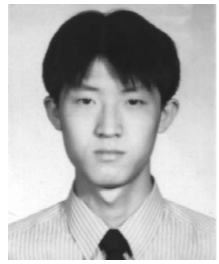

Tao Huang received his BS degree in theoretical physics and MS degree in optics from Shanxi University in 2001 and in 2003, respectively. Now he is pursuing the PhD degree. His research topics concern single-molecule spectroscopy and quantum cryptography.

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